

Definition of a Complete Family

- Suppose $X_1, \dots, X_n \stackrel{iid}{\sim} \text{Poisson}(\theta)$, where $\theta > 0$.
- pdf:

$$f(x_1, \dots, x_n; \theta) = \prod_{i=1}^n \frac{\theta^{x_i} e^{-\theta}}{x_i!} = \frac{\theta^{\sum_{i=1}^n x_i} e^{-n\theta}}{\prod_{i=1}^n x_i!}$$

- $Y_1 = \sum_{i=1}^n X_i$ is a sufficient statistic. $Y_1 \sim \text{Poisson}(n\theta)$ with pdf

$$g_1(y_1; \theta) = \frac{(n\theta)^{y_1} e^{-n\theta}}{y_1!}, \quad y_1 = 0, 1, 2, \dots$$

- Suppose $\mathbb{E}_\theta(u(Y_1)) = 0$ for all $\theta > 0$. Can we get $u(\cdot) = 0$?

$$\begin{aligned} 0 = \mathbb{E}(u(Y_1)) &= \sum_{y_1=0}^{\infty} u(y_1) \frac{(n\theta)^{y_1} e^{-n\theta}}{y_1!} \\ &= u(0) + [nu(1)]\theta + \frac{n^2 u(2)}{2!} \theta^2 + \dots \end{aligned}$$

We have $u(0) = u(1) = \dots = 0$. That is $u(y) = 0$ for $y = 0, 1, 2, \dots$.

- If $\mathbb{E}_\theta[u(Y_1)] = 0$ for all $\theta > 0$, we have $u = 0$ except on a set of points that has probability 0.

- Definition: $Z \sim h(z; \theta)$, $\theta \in \Omega$. If $\mathbb{E}_\theta[u(Z)] = 0$ for all $\theta \in \Omega$, this implies that $u(z)$ is zero except on a set of points with probability zero for each $h(z; \theta)$. Then $\{h(z; \theta) : \theta \in \Omega\}$ is called a complete family.

- $\left\{ \text{Bin}(2; \theta) : 0 < \theta < 1 \right\}$ is a complete family.

- $\{Unif(-\theta, \theta) : \theta > 1\}$ is not a complete family.
- $\{N(0, \theta) : \theta > 0\}$ is not a complete family.